

Sr. Python AI/ML Developer

MD JAHIDUL HASAN

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PROFESSIONAL SUMMARY

Experienced and Sr Results-driven AI/ML & Generative AI Engineer with 10+ Years extensive experience in designing, developing, and deploying scalable machine learning and GenAI solutions using Python across cloud-native environments. Proven expertise in LLMs, Retrieval-Augmented Generation (RAG), MLOps, and data engineering, with strong hands-on experience in AWS, Azure, and GCP platforms. Adept at translating complex business requirements into production-ready AI systems, ensuring model explainability, performance, security, and cost optimization. Experienced in Agile/Scrum environments with a strong focus on automation, reliability, and measurable business impact.

- Designed and delivered end-to-end AI/ML and Generative AI solutions, covering data ingestion, feature engineering, model training, evaluation, deployment, and monitoring.
- Built and optimized machine learning models using Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, and Keras for structured and unstructured data.
- Developed LLM-powered applications using OpenAI, Azure OpenAI, Hugging Face models, integrated through LangChain, LlamaIndex, and Haystack.
- Implemented Retrieval-Augmented Generation (RAG) architectures using FAISS, Pinecone, ChromaDB, Weaviate, and Milvus to enable accurate, context-aware AI responses.
- Performed advanced data analysis and feature engineering using Pandas, NumPy, PySpark, Apache Spark, and Dask.
- Designed and maintained scalable data pipelines using Kafka and Airflow for batch and real-time processing.
- Developed secure and scalable ML/LLM APIs using FastAPI and Flask for real-time inference.
- Deployed and managed ML workloads on AWS (SageMaker, Bedrock, EC2, S3, Lambda, Glue, Athena) and Azure/GCP (Azure ML, Azure OpenAI, Vertex AI, BigQuery).
- Established MLOps pipelines with MLflow, Weights & Biases, DVC, Docker, Kubernetes, enabling model versioning, reproducibility, and CI/CD automation.
- Implemented infrastructure as code (IaC) using Terraform for consistent multi-environment deployments.
- Monitored system health and model performance using Prometheus, Grafana, and cloud-native monitoring tools.
- Created interactive dashboards and visualizations using Matplotlib, Seaborn, Plotly, Power BI, and Tableau to communicate insights to stakeholders.
- Ensured model explainability and compliance using SHAP and LIME, supporting governance and regulatory requirements.
- Optimized model performance, latency, and cost, including LLM token usage and inference efficiency.
- Collaborated with product owners, data engineers, and business stakeholders to define KPIs and deliver AI solutions aligned with business goals.
- Followed Agile/Scrum practices, contributing to sprint planning, reviews, and continuous improvement initiatives

TECHNICAL SKILLS

Programming Languages	Python, SQL, Bash
Machine Learning & AI	Scikit-learn, TensorFlow, PyTorch, Keras, XGBoost, LightGBM
Generative AI / LLMs	OpenAI (GPT-4/3.5), Azure OpenAI, Claude, Gemini, LLaMA, Mistral, Hugging Face Transformers
LLM Frameworks	LangChain, LlamaIndex, Haystack
Fine-Tuning & Optimization	PEFT, LoRA, QLoRA
Data Processing & Analytics	Pandas, NumPy, Dask
Big Data & Streaming	Apache Spark, PySpark, Kafka
Workflow Orchestration	Apache Airflow
Vector Databases (RAG)	FAISS, Pinecone, ChromaDB, Weaviate, Milvus
Databases – SQL	PostgreSQL, MySQL

Databases – NoSQL	MongoDB, DynamoDB, Redis
Model Serving & APIs	FastAPI, Flask, BentoML
ML Platforms / MLOps	MLflow, Weights & Biases, DVC
Containerization & Orchestration	Docker, Kubernetes
Cloud Platforms – AWS	SageMaker, Bedrock, EC2, S3, Lambda, Glue, Athena
Cloud Platforms – Azure/GCP	Azure ML, Azure OpenAI, Vertex AI, BigQuery
Monitoring & Observability	Prometheus, Grafana, CloudWatch
CI/CD & DevOps	Git, GitHub Actions, Jenkins, Terraform
Data Visualization & BI	Matplotlib, Seaborn, Plotly, Power BI, Tableau
Development Tools	JupyterLab, VS Code, Postman, Swagger
Methodologies & Practices	Agile/Scrum, CRISP-DM
Model Explainability	SHAP, LIME

Work Experience

Sr. Python AI/ML Developer || Cigna, Philadelphia, Pennsylvania || Jan 2025 – Current

Roles & Responsibilities:

- Designed, developed, and deployed end-to-end machine learning and AI solutions using Python, following CRISP-DM methodology from data understanding to production deployment.
- Built and optimized ML models using Scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, and Keras for classification, regression, NLP, and time-series use cases.
- Developed Generative AI applications leveraging LLMs (GPT-4/3.5, Claude, Gemini, LLaMA, Mistral) using LangChain, LlamaIndex, and Haystack.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using FAISS, Pinecone, ChromaDB, Weaviate, and Milvus to enable accurate, context-aware LLM responses.
- Applied parameter-efficient fine-tuning (PEFT) techniques such as LoRA and QLoRA to customize large language models while reducing compute costs.
- Performed advanced data preprocessing, feature engineering, and exploratory data analysis (EDA) using Pandas, NumPy, and Dask.
- Designed scalable data pipelines using Apache Spark, PySpark, Kafka, and Airflow for batch and streaming data processing.
- Developed RESTful APIs and inference services using FastAPI and Flask, enabling real-time ML and GenAI model serving.
- Built interactive ML dashboards and prototypes using Streamlit to showcase model insights to business stakeholders.
- Containerized ML applications using Docker and deployed to Kubernetes for scalable and resilient production environments.
- Managed model lifecycle and experimentation using MLflow and Weights & Biases, including versioning, tracking, and artifact management.
- Deployed and managed ML workloads on AWS (SageMaker, Bedrock, EC2, S3, Lambda, Glue, Athena) and Azure/GCP (Azure ML, Azure OpenAI, Vertex AI, BigQuery).
- Designed and optimized SQL and NoSQL databases including PostgreSQL, MySQL, MongoDB, Redis, DynamoDB for ML applications.
- Implemented CI/CD pipelines using Git, GitHub Actions, Jenkins, Terraform, enabling automated testing, model deployment, and infrastructure provisioning.
- Monitored model performance and system health using Prometheus and Grafana, ensuring reliability and SLA compliance.
- Created insightful data visualizations using Matplotlib, Seaborn, Plotly, Power BI, and Tableau for analytics and executive reporting.
- Ensured model explainability and transparency using SHAP and LIME to meet compliance and business interpretability requirements.
- Collaborated in Agile/Scrum teams, participating in sprint planning, code reviews, and cross-functional stakeholder

discussions.

- Used JupyterLab, VS Code, Postman, Swagger, and DVC for experimentation, API testing, documentation, and data version control.
- AWS Fargate created a orchestration service to deploy and manage Docker containers.
- Create END-END System development with Modernization in AWS.
- Integrated Jenkins, TeamCity, Octopus, and Ansible with other DevOps and development tools, such as Git, Docker, and AWS, creating end-to-end automation workflows.
- With DML Developed complex SQL queries for retrieve data from multiple tables in a database schema, resulting in improved data analysis and reporting capabilities to better visibility for business to understand field needs
- Collaborated with cross-functional teams, including Salesforce developers and administrators, to design and implement seamless integration workflows between Snowflake and Salesforce applications.
- Created web mapping applications using ArcGIS tools with Spotfire.
- Involved in developing micro services which are interconnected in the AWS cloud that involved in consuming and building web services with both SOAP and RESTful.
- Developed Restful Micro services using Flask and Django and deployed on AWS servers through EC2 and Other machine learning models, including support vector machine.
- Used Web Pack, Babel and gulp for transpilation (compilation) configuration for Typescript to be converted to JavaScript.
- Developing test automation framework using python and See Test APIs.
- Genetic algorithms and heuristic modeling for non-differentiable problems, Cluster analysis and unsupervised learning.
- Implemented various screens for the front end using React.js and used various predefined components from NPM (Node Package Manager) and redux library.
- Developed front-end using Html, CSS, JavaScript, React, Redux, Boot strap.
- Conduct root cause analysis and complex performance tuning. Wrote queries and designing schemas.
- Knowledge of popular Scala frameworks and libraries, such as Akka, Play, and Spark, for building reactive applications and processing large datasets.
- Experience in designing and implementing distributed systems using Scala, ensuring scalability, fault tolerance, and efficient data processing.
- Developed new Widget-app module for MMD 2.0 using React. js with Redux architecture and ES6.
- AWS EAF with service web application common web exploits and vulnerabilities with rules to block or allow incoming traffic based on IP addresses, headers, and content with geospatial technologies ArcGIS development.
- With Data Science Concepts implemented and conducted data regression analyses of the relationship between Explosion scenario and grid error, achieving a 8.3% more accurate prediction of performance than previous application.
- Utilize web scraping techniques to extract and organize competitor data Update risk analysis techniques such as data recall and segmentation, resulting in a 14% increase in usability.
- PySpark integrated systems with support data processing workflows with data lake Worked with DDoS Shield team to mitigate and service disruptions

Environment: C, C++, Python, SQL, PL/SQL, EDI, UNIX, Linux, Shell Scripting, Solaris, MMIS, Sybase, Perl Scripting, UML, JSON.

Python AI/ML Developer || Morgan Stanley, New York, United States || Sept 2023- Dec 2024

Roles & Responsibilities:

- Architected production-grade LLM workflows integrating prompt engineering, context management, and memory using LangChain and LlamaIndex.
- Designed enterprise-scale RAG systems combining vector databases and structured data sources to improve response accuracy and reduce hallucinations.
- Optimized model inference latency and cost through batching, quantization, caching, and GPU utilization strategies.
- Built event-driven ML pipelines using Kafka and AWS Lambda for near real-time data ingestion and inference.
- Automated data validation, feature checks, and pipeline monitoring to ensure data quality and model reliability.
- Implemented A/B testing and champion-challenger models to evaluate model performance in production.
- Developed secure LLM APIs with authentication, authorization, and rate limiting using FastAPI and cloud-native

services.

- Integrated OpenAI, Azure OpenAI, and Hugging Face models with internal enterprise systems and databases.
- Performed hyperparameter tuning and model optimization using distributed compute frameworks such as Spark and Ray.
- Established model governance and version control standards using MLflow, DVC, and Git, ensuring reproducibility and auditability.
- Designed scalable cloud infrastructure using Terraform for ML workloads across AWS and Azure environments.
- Implemented logging, tracing, and alerting for ML services using Prometheus, Grafana, and CloudWatch.
- Developed custom embeddings and semantic search solutions for document search, chatbots, and knowledge assistants.
- Collaborated with product and business teams to translate business requirements into ML/AI solutions with measurable KPIs.
- Conducted bias detection, fairness analysis, and explainability assessments to meet ethical AI and compliance standards.
- Led code reviews and technical mentoring for junior ML engineers and data scientists.
- Created technical documentation and API specifications using Swagger/OpenAPI for ML and GenAI services.
- Delivered POCs and MVPs for GenAI use cases and successfully scaled them into production systems.
- Supported multi-cloud deployments ensuring portability and resilience across AWS, Azure, and GCP.
- Participated in incident response and root-cause analysis for ML system failures and performance degradation.
- Experienced of mounting Azure file share by using CIF.
- Working Experience on Azure Storage, SQL Azure and in different PaaS Solutions with Web, and worker Roles and Azure Web Apps.
- Designed and maintained databases using Python and developed Python-based API (RESTful Web service) using Flask and Postgre SQL.
- Actively involved in Initial software development life cycle (SDLC) of requirement gathering and in suggesting system configuration specifications during client interaction.
- Created Python and Bash tools to increase efficiency of call center application system and operations; data conversion scripts, AMQP/Rabbit MQ, REST, JSON, and CRUD scripts for API Integration.
- Used Celery with Rabbit MQ, MySQL, Elastic search, and Flask to create a distributed worker framework.
- Consumed REST based Micro services with Rest template based on RESTful APIs and designed, developed and tested HTML, CSS, jQuery that meets accessibility and web browser standards.
- Develop Python micro services with Django/Flask framework for confidential internal web applications.
- Used Docker services for deployment and testing of micro services.
- Designed, created, executed, and managed Docker, Jenkins, and Git for micro services.
- Automated Rabbit MQ cluster installations and configuration using Python/Bash.
- Used Python Library BeautifulSoup for web scrapping to extract data for building graphs.
- Used object-relational mapper (ORM) to automate the transfer of data stored in relational databases tables into objects.
- Created Power BI Reports using the tabular data models as source data in BI desktop and publish reports to service.
- Designed and developed web crawler in python using scrapy framework and using Rabbit MQ as a messaging server between the micro services.
- Developed Kafka producer and consumers for real time high traffic software.
- Worked on Angular framework to develop interactive websites based on client needs.
- Used Jenkins pipelines to drive all micro services builds out to the Docker registry and then deployed to Kubernetes, created pods and managed using Kubernetes.

Environment: Python, Django Web Framework, Flask, RESTful API, Kafka, Digital Ocean, HTML, CSS, JavaScript, JSON, Ajax, MySQL, Postgre SQL, Sublime Text, Atom, PyCharm, Docker, Jenkins, Kubernetes GIT, Unit test, Locust.

Python AI/ML Developer || Flagstar Bank, Troy, Michigan || Aug 2022- Aug 2023

Roles & Responsibilities:

- Engineered prompt orchestration frameworks with versioned prompts, evaluation metrics, and rollback strategies for enterprise LLM applications.
- Built hybrid search solutions combining keyword, semantic, and vector search to enhance enterprise document

discovery.

- Implemented streaming inference pipelines using Kafka and Spark Structured Streaming for real-time ML and GenAI use cases.
- Designed multi-tenant ML platforms supporting isolated models, datasets, and access controls across business units.
- Developed cost-optimization strategies for LLM usage, including token budgeting, prompt compression, and intelligent routing across models.
- Created custom evaluation frameworks to measure LLM performance using accuracy, relevance, toxicity, and hallucination metrics.
- Implemented secure data masking and PII redaction within AI pipelines to ensure regulatory compliance.
- Integrated feature stores and reusable data assets to standardize ML feature engineering across teams.
- Designed high-availability ML inference architectures with autoscaling and failover on Kubernetes.
- Built model drift detection and automated retraining pipelines based on data and prediction drift signals.
- Automated ML environment provisioning using Infrastructure as Code (IaC) across dev, QA, and prod.
- Developed serverless ML inference workflows using AWS Lambda and cloud-native orchestration.
- Implemented role-based access control (RBAC) and secrets management for ML and GenAI systems.
- Conducted performance benchmarking across multiple ML and LLM models to select optimal solutions.
- Designed AI observability dashboards tracking latency, cost, accuracy, and system health.
- Led cross-functional design reviews to align AI solutions with enterprise architecture standards.
- Built knowledge assistants and chatbots integrated with enterprise data, APIs, and workflows.
- Established best practices for responsible AI, including transparency, fairness, and auditability.
- Migrated legacy analytics workflows to cloud-native ML platforms improving scalability and reliability.
- Supported production readiness reviews and go-live activities for AI/ML solutions.
- Worked on reading and writing multiple data formats like JSON, ORC, Parquet on HDFS using PySpark.
- Developed internal auxiliary web apps using Python - Flask framework with HTML/CSS.
- Created entire application using Python, Django, MySQL and Linux.
- Utilized PyUnit the Python unit test framework, for all Python applications.
- Gained Knowledge on Deploying apps using AWS Cloud Formation.
- Developed views and templates with Python and Django's view controller and templating language to create a user-friendly website interface.
- Troubleshoot and provide support/monitoring reports in BI workspaces whenever there are issues in production.
- Working closely with DevOps team for developing efficient deployment mechanisms.

Environment: Python, Django, Flask, HTML5/CSS, UNIX, MS SQL Server, AWS, BI Reports, JavaScript, AJAX, jQuery, Angular, Selenium, Eclipse, Linux, Shell Scripting, Restful, MVC3.

Python Developer || NISUM, Hyderabad, India || May 2017 – June 2021

Roles & Responsibilities:

- Involved in Web application penetration testing process, web crawling process to detect and exploit SQL Injections Vulnerabilities.
- Wrote Test Automation Script for deployment of AWS and Company Web API, using python package, puppet configuration tool to update and check historical pumping data from AWS. Compare the existing data and historical data and create performance chart for client.
- Wrote automate Python Script for testing program to store machine detection alarm when Pump experience overloading to Amazon cloud.
- Wrote program to fetch data and from amazon cloud and send the client according their requirement.
- Passed query and wrote script to automate machine alarm data with timestamp to engineer.
- Used Python and Pandas library, built data analysis graph for documentation and record.
- Worked with Blender script, built add-on and customization of blender interface.
- Used SPSS tool for statistical programming and computational techniques for large data sets and quantitative analysis.
- Used SPSS tool for Visualization techniques for summarizing complex data analysis using Python.
- Independently developed and implemented analysis workflow and quality practices, including workflow optimization, process standardization, configuration management of analysis files and software, and verification and validation (V&V) of models.
- Using Numerical equation, built application and do 2D- finite Element analysis using python language

- Used Django database API's to access database objects.
- Generated graphical reports using python package Numpy and Matplotlib.
- Representation of the system in hierarchy form by defining the components, subcomponents using Python and developed set of library functions over the system based on the user needs.
- Development of Python APIs to dump the array structures in the Processor at the failure point for debugging.
- Extracted the actual data of HTML format and predicted raw data; interpreted and stored predicted one in well-organized JSON files.
- Wrote programs to parse excel file and process many user data with data validations.
- Used Python and Django to interface with the JQuery UI and manage the storage and deletion of content.
- Maintained data with different levels of access privileges.

Environment: Python2.7, Amazon Cloud, Blender, C#, Macros Oracle DB, Debian, Apache Server, pandas Django, MySQL, Linux, HTML, GIT, CSS, JavaScript.

Software Developer || Metamor Software Solutions, Hyderabad, India || Sep 2015 – Apr 2017

Roles & Responsibilities:

- Participated in requirement gathering and worked closely with the architect in designing and modeling.
- Used Python and Django Framework to develop the web application's backend and REST API.
- Used Django and manual SQL modifications to modify Django models while retaining all data while the site was in production mode, Implemented database access using Django ORM.
- Implemented URL parser to display markers on the user interface for images using JSON, JQuery, and JavaScript.
- Collaborated with business stakeholders to understand their data requirements and created customized Tableau dashboards to address their specific needs.
- Transformed raw data into meaningful insights using Tableau visualizations, enabling cross-functional teams to identify trends and opportunities.
- Developed interactive Tableau dashboards to showcase key performance metrics, resulting in improved data accessibility and faster decision-making.
- Wrote Python routines to log into the websites and fetch data for selected options, Worked on Python Modules and Packages.
- Wrote Python scripts to parse JSON documents and load the data in the database.
- Created a database using MySQL and wrote several queries to extract data from the database.
- Responsible for analyzing various cross-functional, multi-platform applications systems enforcing Python best practices and guiding in making long-term, scalable architectural design decisions.
- Provide support during implementation and coordinate with the production team.

Environment: Python, Django, JSON, JQuery, JavaScript, XML, MySQL.